

Predicting Marriage Compatibility Using TabTransformer

Final Report

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Github: <https://github.com/YanmiYu/DL-Marriage-Prediction/tree/main>

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1. Introduction

This project investigates how **structured personal attributes** influence **romantic partner preferences**, using 9,160 wedding announcements from *The New York Times* (2013–2023). Each wedding announcement article contains demographic profiles—such as gender, age, education, and occupation—for both individuals and their parents, extracted via GPT 3.5 turbo and standardized via GPT-4o mini preprocessing.

Combining insights from sociological studies and advances in deep learning for tabular data, we aim to **model social pairing patterns** as a multi-label, multi-class classification task: given one individual’s profile, the model predicts key attributes of their partner (e.g., gender, age group, education, and occupation).

Through this architecture, we seek not only to explore potential patterns in matchmaking behavior but also to understand how machine learning might **reflect—or reinforce—existing social stratification**. Marriage data encodes cultural preferences and elite norms, and our model helps surface these latent structures. We hope this approach bridges computational methods and sociological inquiry, offering both predictive insight and pathways for fairer, more interpretable modeling.

1.1 Related Work: TabTransformer

In exploring neural architectures for structured data, we focused on the **TabTransformer** (Huang et al., 2020), a deep learning model that applies self-attention to learn contextual embeddings for categorical features. Unlike standard MLPs, which treat each feature independently, TabTransformer encodes dependencies between columns, for example, how education may relate to occupation or location, through Transformer layers. This improves generalization and robustness, particularly in settings with noisy or missing data.

To contextualize our model choice, we also reviewed **TabNet** (Arik & Pfister, 2021), which uses a sequential attention mechanism to select feature subsets at each decision step. While TabNet emphasizes interpretability and adaptive feature selection, we prioritized TabTransformer’s ability to model **column-wise interactions** through attention, which aligns well with our task of inferring partner traits from structured profiles.

In our implementation, we adapted the original TabTransformer architecture to handle partner attribute prediction, using column-specific embeddings and shared attention blocks to jointly model features like age, gender, education, and occupation. This design allowed the model to capture structured patterns in the data more effectively.

2. Data Preprocess

Our dataset builds on preprocessed wedding announcements collected by Dr. Zhenchao Qian and Guixing Wei’s research group at Brown University, which used the NYT Article Search API to identify relevant articles (2013–2023) via keywords ("wedding," "vow") and temporal heuristics (e.g., section tags). Their team web scraped raw text, retained metadata, and extracted 60 demographic information via ChatGPT 3.5 turbo.

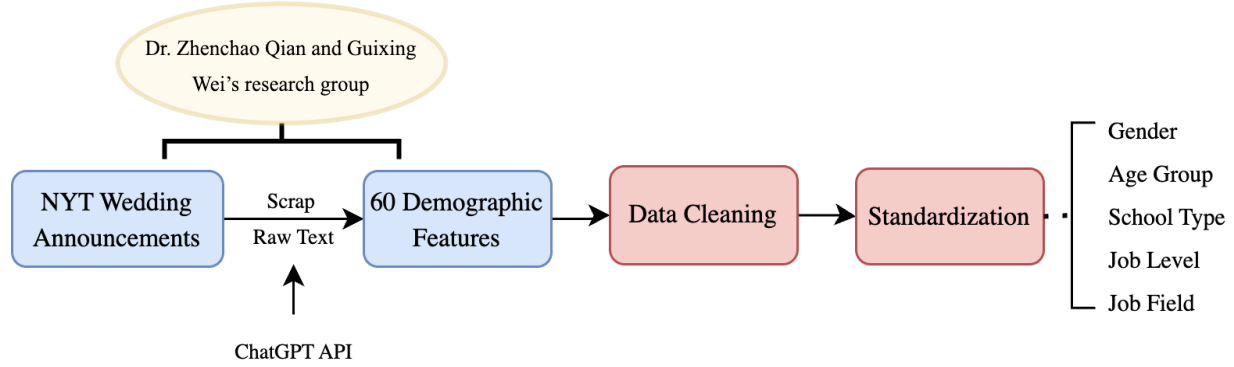


Figure 1. Data Preprocess Pipeline

We further processed this dataset through **multi-stage normalization**, applying rule-based and statistical methods to clean and categorize fields for modeling. Firstly, occupations were mapped along two dimensions: job seniority (e.g., Entry Level to Executive), based on LinkedIn’s hierarchy, and broad industry categories (e.g., Business and Financial Occupations, Media and Communication Occupations) according to Bureau of Labor statistics, and refined using the ChatGPT 4o-mini batch API. Secondly, undergraduate institutions were grouped into five tiers (e.g., Ivy League, Top 50 Public), along with “No College Mentioned” used when data was missing. Graduate school attendance was encoded as a binary feature. Lastly, age was extracted or inferred from the raw data, then binned into five-year intervals; gender was standardized into four categories (Male, Female, Nonbinary, Unknown). All categorical fields were label-encoded, with “Unknown” consistently used for missing or unextractable values. This pipeline allowed us to transform unstructured wedding announcement text into a **rich, structured dataset of partner attributes** suitable for supervised learning via **deep tabular models**.

3. Model Structure

We developed an **ImprovedTabTransformer** for structured demographic prediction, using transformer-based attention to model dependencies across five categorical input fields (e.g., gender, age, education). Each field is embedded into a 32-dim vector and combined with a

learnable per-feature column embedding, then concatenated with a learnable [CLS] token but omits positional encoding due to fixed field order. The idea of CLS token is from BERT. The sequence is processed by a 4-layer Transformer encoder (8 heads, 256 hidden units, dropout=0.4) with batch-first processing.

The [CLS] token's output feeds into five deep task-specific MLP heads (each with 3 linear layers, GELU activations, and LayerNorm) to jointly predict partner attributes: gender, age group, school category, job level, and job field. This multi-task setup enables shared representation learning while maintaining task specialization.

To establish a baseline, we implemented a simple **MLP** that takes integer-encoded categorical features as a flattened input (no embeddings). It consists of two shared ReLU-activated layers (128 hidden units), followed by five parallel linear heads for predicting partner attributes. To improve robustness, inputs are randomly masked with 20% probability during training. The model is optimized with Adam (lr=0.001). Despite lacking attention or field-aware structure, this baseline offers a clean benchmark against the TabTransformer.

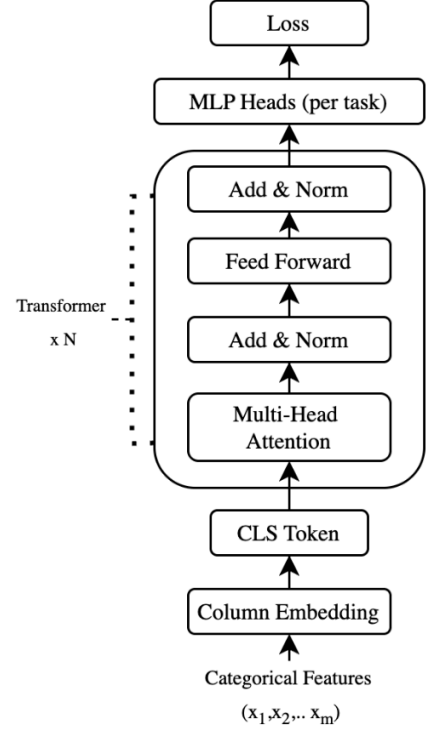


Figure 2. Model Architecture

4. Results

We evaluate our models on five attribute prediction tasks: gender, age group, undergraduate school category, job level, and work field, with unweighted. Performance is measured by classification **accuracy** for each output head on the held-out test set.

$$\text{Accuracy}_j = \frac{1}{N} \sum_{i=1}^N \left[\hat{y}_i^{(j)} = y_i^{(j)} \right], \quad \left[\hat{y}_i^{(j)} = y_i^{(j)} \right] = \begin{cases} 1, & \text{if } \hat{y}_i^{(j)} = y_i^{(j)} \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Average Accuracy} = \frac{1}{K} \sum_{j=1}^K \text{Accuracy}_j$$

Our Improved TabTransformer achieves an average accuracy of 55%, outperforming the MLP baseline's 44%. As shown in Figure 4, the TabTransformer performs best on gender (84%), followed by age group (63%) and school category (51%). Performance is lower on

occupation-related tasks—job level (45%) and job field (35%)—due to the higher variability and ambiguity of these features.

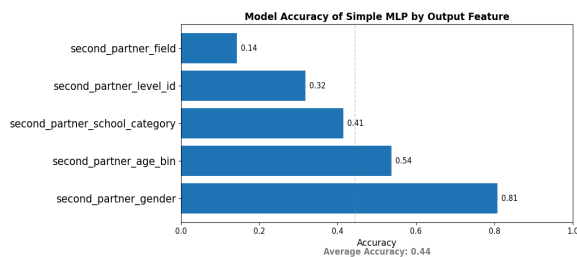


Figure 3. Model Accuracy of Simple MLP

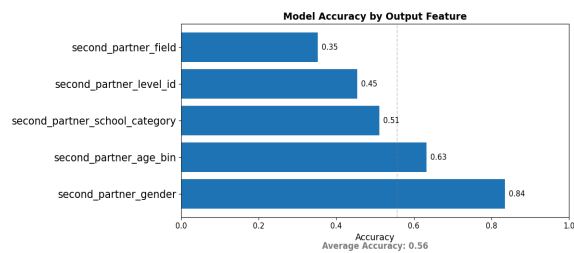


Figure 4. Model Accuracy of Tabtransform

As shown in Figure 3 and 4, comparing the performance between TabTransformer and a simple multilayer perceptron (MLP) baseline with the same input/output structure, our TabTransformer **consistently outperforms** the MLP across all tasks, with especially notable improvements on job field prediction (35% vs 14%, increased by 21%). This suggests that TabTransformer's attention-based structure is better suited to **modeling sparse and noisy features** like occupation. Even on well-structured features such as age and gender, the transformer yields modest improvements (e.g., 84% vs. 81% on gender).

In addition to quantitative results, we also include a sample **qualitative prediction**. Given a female individual aged 30–34 from an Ivy League background working in Computer and IT, the model predicts a male partner with a similar age and educational background, but a shifted occupational domain (e.g., Business & Finance), illustrating how the model captures plausible social pairing patterns while allowing for variance across dimensions.

Table 1. Sample Prediction

Attribute	Input Partner → Predicted Partner
Gender	Female → Male
Age Group	30-34 → 30-34
Undergrad Category	Ivy League → Ivy League
Job Level	S4 → S4
Field of Work	Computer and Information Technology Occupations → Business and Financial Occupations

5. Challenge

Throughout the project, we encountered a variety of technical and conceptual challenges spanning data extraction, model design, and broader socio-technical limitations.

For **data preprocessing and extraction**, initially, early attempts to standardize occupations using a fixed list of 343 Bureau of Labor titles via the ChatGPT API produced inconsistent results—GPT often returned unmatched or incorrect labels. Keyword search, on the other hand, proved inadequate due to the wide variety of job titles. We addressed this by simplifying the task: mapping occupations to two standardized variables—**job level** and **field of**

work—each assigned a **unique ID**. We then prompted GPT to return only the corresponding ID, which significantly improved **consistency**, **scalability**, and **output quality**.

To handle **ambiguous fields**, we applied **data normalization** strategies especially for schools and occupation. For example, vague job titles like “analyst” were clarified using employer context, and school aliases (e.g., “UPenn”) were mapped to canonical names. Despite these efforts, some categories such as occupation remained sparse and noisy, contributing to class imbalance.

On the **modeling side**, building a multi-output prediction pipeline with TabTransformer involved significant engineering, including task-specific heads and joint loss functions. Additionally, the NYT dataset itself reflects structural bias and features mostly elite, highly-educated East Coast couples, which constrain the generalization over larger populations.

6. Discussion & Reflection

6.1 Discussion

While including 9,160 wedding announcements from *The New York Times* (2013–2023) with rich information, our dataset is highly selective and not representative of the general population. The data reflects both **self-selection bias**, since it only includes couples who are aware of the NYT wedding section and choose to submit their announcements, and **editorial bias**, as the NYT tends to publish stories that highlight affluent, highly educated individuals. As a result, the dataset disproportionately features **elite, white, cisgender couples**, limiting its demographic diversity and the generalizability of model outcomes beyond this sociocultural context.

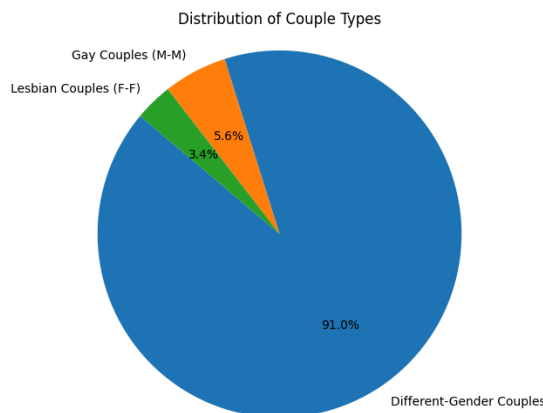


Figure 5. Distribution of Couple Types in Dataset

Moreover, while we include four gender classes (Male, Female, Nonbinary, Unknown), **nonbinary representation is extremely sparse**, and **same-gender couples are overrepresented** (~10%) relative to national statistics (~1%). This imbalance, possibly driven by visibility or editorial emphasis, complicates fairness analysis. Our dataset includes approximately 91% heterosexual couples, 5.6% gay couples, and 3.4% lesbian couples, as shown in Figure 5. Other LGBTQ+ identities are not modeled due to a lack of structured data.

Overall, the model may learn patterns embedded in **elite social norms**, such as educational homogamy, rather than universal matchmaking trends, and results should be interpreted within this biased, socially constrained framework. These limitations underscore the need to critically examine how models trained on biased datasets may **reinforce**, **distort**, or merely **reflect** existing social inequalities. Responsible use of such models requires transparency about these biases and caution in interpreting or deploying predictions in broader contexts.

6.2 Reflection

We successfully met our **base goal** of building a functional model to predict partner attributes from individual profiles. We also achieved our **target goal** by extending the model to output the top 3 or top k most likely bundles of ideal partner characteristics(see Table 2).

Rank	Attribute	Prediction (Score)
1	Gender	Male (0.76)
	Age Group	30–34 (0.64)
	School Category	Other Colleges (0.47)
	Education Level	S4 (0.46)
	Field of Study	Business and Financial (0.19)
2	Gender	Male (0.76)
	Age Group	30–34 (0.64)
	School Category	Other Colleges (0.47)
	Education Level	S4 (0.46)
	Field of Study	Arts and Design (0.17)
3	Gender	Male (0.76)
	Age Group	30–34 (0.64)
	School Category	Other Colleges (0.47)
	Education Level	S3 (0.20)
	Field of Study	Media and Communication (0.15)

Table 2. Top k Predicted Bundles of Ideal Partner Attributes (k =3)

Although we only partially met our **stretch goal**—we were able to assign compatibility scores between individuals but did not collect new data from alternative sources or years—our current results demonstrate strong potential(see Table 3).

Candidate	Match %	Gender (Score)	Age Group (Score)	School Category (Score)	Education Level (Score)	Field (Score)
1	52.0%	Male (0.73)	30–34 (0.68)	Ivy League (0.28)	S4 (0.43)	Legal (0.12)
2	34.8%	Female (0.21)	30–34 (0.68)	Top 50 Private (0.23)	S4 (0.43)	IT (0.06)
3	28.6%	Male (0.73)	25–29 (0.14)	Top 30 Public (0.05)	S3 (0.13)	Education (0.05)

Table 3. Compatibility Score Between the Target individual and k Candidate Profiles ($k = 3$)

As anticipated, the model performed best on well-defined, less ambiguous features, which reinforced our initial hypothesis that attention-based architectures are especially effective in **handling noisy and interdependent data**.

Over time, our approach evolved. On the **model side**, we initially experimented with the original TabTransformer architecture but found the results suboptimal for our prediction tasks. To improve performance, we introduced a learnable [CLS] token inspired by BERT, concatenated to the sequence of embedded categorical fields. We omitted positional encoding due to the fixed order of input fields. On the **data side**, we started with a basic keyword-matching approach and later transitioned to a ChatGPT batch API pipeline, which we iteratively refined to improve the consistency and structure of attribute extraction. In hindsight, we would invest earlier in data validation tooling and fairness diagnostics.

If we could do the project over again, we would invest earlier in **data validation tools** and **fairness diagnostics**, as these became increasingly important during model evaluation. With more time, we would also explore longitudinal trends, apply soft or hierarchical labeling to ambiguous categories, conduct more thorough **hyperparameter tuning**, and seek **alternative data sources** to increase diversity and generalizability.

Ultimately, we learned that model architecture alone is not enough—**data quality and framing** significantly impact outcomes. This project deepened our understanding of working with messy, real-world data and reminded us that **fairness, transparency, and structure** must be considered alongside predictive accuracy.

7. References

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